

Enhanced diabetic retinopathy detection using U-shaped network and capsule network-driven deep learning^{☆,☆☆}

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ABSTRACT

Glaucoma, a severe eye disease leading to irreversible vision loss if untreated, remains a significant challenge in healthcare due to the complexity of its detection. Traditional methods rely on clinical examinations of fundus images, assessing features like optic cup and disc sizes, rim thickness, and other ocular deformities. Recent advancements in artificial intelligence have introduced new opportunities for enhancing glaucoma detection. This research explores a hybrid approach combining UNet++ and Capsule Network (CapsNet) architectures for accurate glaucoma diagnosis. UNet++ is employed for semantic segmentation, focusing on defining optic discs and cups, which are crucial for detecting the disease. CapsNet leverages its ability to recognize hierarchical patterns, providing more sensitive detection of glaucomatous changes than conventional Convolutional Neural Networks. Pre-processing of retinal images involves advanced techniques like Histogram Equalization and Contrast Limited Adaptive Histogram Equalization (CLAHE) to enhance image quality. The model is trained and tested on benchmark datasets, showing superior performance in optic cup/disc segmentation and glaucoma detection accuracy compared to existing state-of-the-art models.

- **Hybrid Model Efficiency:** The combined use of UNet++ and CapsNet offers improved accuracy in optic cup and disc segmentation.
- **Enhanced Image Quality:** Application of Histogram Equalization and CLAHE techniques significantly boosts the quality of retinal images.
- **Superior Performance:** The hybrid approach outperforms traditional and contemporary models in glaucoma detection accuracy.

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Resource availability:	1. ORIGA-Light Dataset: Annotated fundus images for optic disc and optic cup segmentation. URL: ORIGA-Light Dataset 2. Software: TensorFlow/Keras: A widely used deep learning framework for implementing UNet++ and CapsNet models. URL: TensorFlow OpenCV: Open-source library for computer vision tasks, useful for image pre-processing. URL: OpenCV 3. Hardware: NVIDIA GPU: A GPU with CUDA support is recommended for training deep learning models efficiently. URL: NVIDIA CUDA GPUs 4. Pre-processing Techniques: Histogram Equalization and CLAHE Implementation: Available as functions within OpenCV or Python libraries such as skimage. URL: CLAHE in OpenCV

Background

Motivation behind the hybrid UNet++-CapsNet methodology for glaucoma detection

Glaucoma is one of the leading causes of irreversible blindness worldwide. Early detection and treatment are crucial to preventing significant vision loss, yet the disease often goes undetected until it has progressed to an advanced stage [1,2]. Traditional diagnostic methods rely heavily on manual clinical examinations of fundus images, where ophthalmologists assess specific ocular features, such as the optic cup-to-disc ratio, rim thickness, and other structural deformities [3]. However, these methods can be subjective, time-consuming, and require a high level of expertise, leading to potential misdiagnoses.

Recent advancements in artificial intelligence (AI) and deep learning have presented new opportunities for improving the accuracy and efficiency of glaucoma detection [4]. AI-driven models have the potential to automate the diagnostic process, providing consistent, rapid, and accurate assessments. Among the various deep learning architectures, Convolutional Neural Networks (CNNs) have been widely adopted for medical image analysis due to their ability to extract complex features from images [5]. However, CNNs have limitations, particularly in capturing spatial hierarchies and preserving detailed structural information, which are critical for accurately detecting subtle glaucomatous changes in retinal images.

This research introduces a hybrid approach that combines the strengths of UNet++ and Capsule Networks (CapsNet) to address these challenges. UNet++, a nested U-Net architecture, is well-suited for semantic segmentation tasks, allowing for precise delineation of the optic disc and cup, which are essential for calculating the cup-to-disc ratio—a key indicator of glaucoma [6]. By leveraging UNet++ for segmentation, the model can more accurately identify and separate the relevant ocular structures from the rest of the image.

Capsule Networks, on the other hand, offer a novel approach to deep learning by encapsulating spatial relationships between features. Unlike traditional CNNs, CapsNet can capture and preserve hierarchical patterns within the image, making it more sensitive to the subtle structural changes associated with glaucoma. By integrating CapsNet into the hybrid model, we aim to enhance the detection of glaucomatous patterns that might be overlooked by conventional CNNs [7].

The motivation behind developing this hybrid UNet++-CapsNet methodology stems from the need for a more robust and reliable automated diagnostic tool for glaucoma. This tool is intended to assist ophthalmologists in making more accurate and timely diagnoses, ultimately reducing the burden of blindness caused by late detection [8]. The methodology is particularly relevant in regions with limited access to specialized healthcare, where early detection through automated systems could make a significant impact [9,10].

Furthermore, this approach is designed to be adaptable and scalable, making it suitable for application in various clinical settings. By improving the accuracy of glaucoma detection, the hybrid model has the potential to enhance patient outcomes and reduce the global burden of glaucoma-related blindness. The integration of advanced image pre-processing techniques, such as Histogram Equalization and CLAHE, further enhances the model’s performance by improving the quality of the retinal images used for analysis.

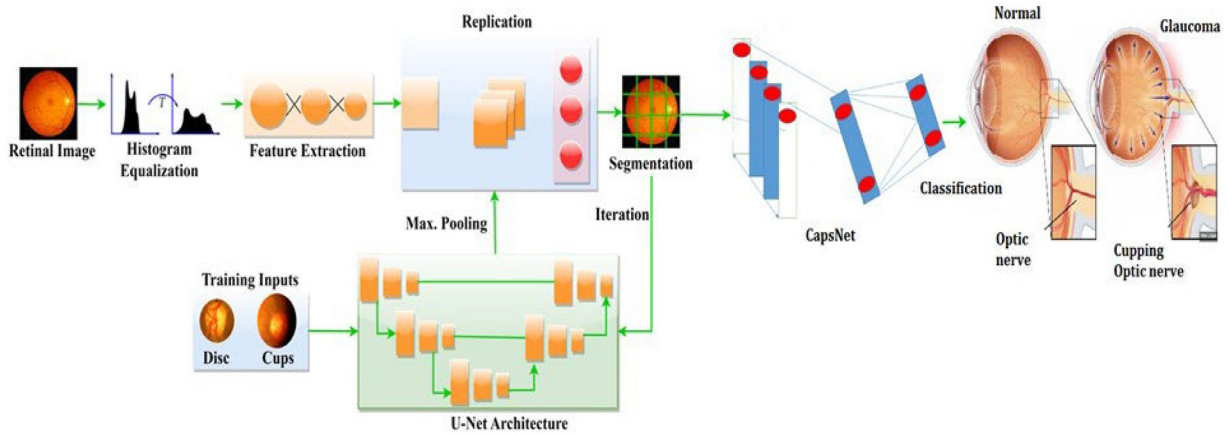


Fig. 1. Illustration of the proposed DRD-CN-DL.

In summary, the hybrid UNet++-CapsNet methodology [21] represents a significant advancement in the field of ophthalmic diagnostics, providing a powerful tool for the early and accurate detection of glaucoma. This research is motivated by the potential to revolutionize glaucoma screening and diagnosis, making high-quality eye care more accessible and effective worldwide.

Method details

The proposed retinal image quality assessment through CNN-based histogram equalization is performed to improve the accuracy and precision of Diabetic Retinopathy Detection (DRD) using Deep Learning (DL). In this, a Capsule network-aided Deep Learning (CN-DL) is implemented for precise feature classification-based DRD purposes. The exact quality of the retinal image is calculated based on the maximum letter size/ combination of the object's physical size and distance from the principle point to the nodal point which is employed for the precise feature extraction. The proposed method with its processes is illustrated in Fig. 1.

From the instance, the actual retinal image quality is computed using Capsule Network and is explained using two convolutional layers with varying textural features. Max pooling features and replicated features are classified here based on extracted features. Next max pooling features are added with 2×2 strides and two or more convolution layers using CN for accurate segmentation. Finally, a max pooling feature with 2×2 strides is used to analyze the segmentation accuracy. The classification of two output layers is used for precise image segmentation with less replicated features.

Pre-Processing of retinal images

In this retinal image pre-processing, the radiologists provide labeling to the initial retinal images for accurate feature classification. In this case, the poor-quality images are identified with the help of rescaling contrast and intensity from extracted features. For instance, the binary operations are assigned for reducing small objects with F1-Score ranges from 2 to 4 between the input retinal images. The retina image pre-processing plays a major role in implementing the Capsule network for diabetic retinopathy detection.

Post pre-processing, a small-scale dataset with a set of retinal images is employed for the diabetic retinopathy detection method. The true purpose of this suggested approach is to identify the quality of retinal pictures using fundus imaging. The retinal images are processed using the suggested U-NET architecture for accurate image segmentation. It matches the training images and segmented images for classifying the replicated input and max pooling input independently using U-NET architecture.

Training U-net architecture

U-Net was implemented in this proposed model for accurate feature-based image segmentation through enlarge and reduce paths. U-Net makes use of performing the cross-connections from enlarge to reduce the path to achieve a more precise image classification output.

For instance, each encoder unit consists of 2-convolutional layers with max-pooling features. At each pooling operation, the accumulated features are doubled. The middle part between the enlarged and reduced path is called as bottleneck. This bottleneck consists of two convolution layers (i.e.) Up convolution layer (Up-sampling) and Down convolution layer (Down-sampling). In the decoder unit, 4 decoders are used to form the reducing path of U-Net. Each decoder unit includes both a de-convolution layer and chaining with the corresponding trimmed feature using the enlarged path [17,18].

U-Net architecture is mainly used for retinal image quality analysis using fundus photography. This architecture performs diabetic retinopathy detection based on corresponding pre-processed images and without increasing the replicated features for further image quality assessment. The fundus photography observed from the human eyes is used to pursue feature classification through the Capsule network.

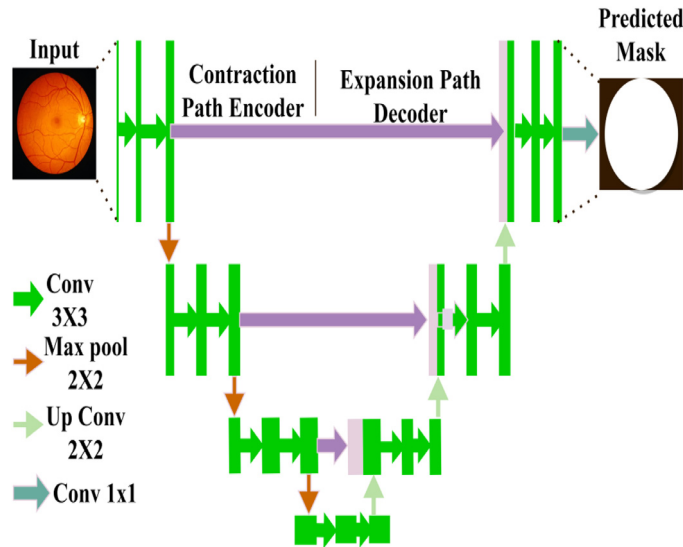


Fig. 2. U-net architecture used for training.

The U-Net architecture is applied to identify and classify the max pooling and replicating features from the given datasets for accurate diabetic retinopathy detection. A feature classification is performed to identify the key factors from input retinal images which reduce the chances of latency during histogram equalization. The U-Net architecture used for training is illustrated in Fig. 2.

The standard U-Net design is made to identify the ROI in Retinal images. The proposed U-NET architecture contains limited information with high detection accuracy. The Capsule network is used in the U-NET architecture which minimizes the computational cost of diabetic retinopathy detection. The U-NET is applied to reduce the classification time and thereby increase the detection accuracy. The proposed architecture uses different features to increase the pooling layer levels.

In the proposed U-NET architecture, Downsampling is referred to as an enlarging path whereas Upsampling is referred to as a reducing path. In the downsampling, the actual size of the retinal images is reduced with increasing training images. The accumulated features are doubled at the time of Downsampling.

The enlarging path is worked by redoing 3×3 convolution layers based on max pooling with kernel size 2×2 and rectified linear unit (ReLU) for precise diabetic retinopathy detection. In this Upsampling, the initial retinal images are enhanced by minimizing the replicated features in input images. It contains the ReLU activation function and 3×3 convolution layers for precise feature classification. Finally, the 1×1 convolution layer is employed in the final layer to find the prominent features for image segmentation [19].

The proposed U-NET architecture is useful in providing training to the replicated feature-identified images through initial retinal images with minimum era values. U-NET architecture improves the diabetic retinopathy detection accuracy and reduces the computational cost. This architecture plays a key role in maximizing the robustness range in the diabetic retinopathy detection process. This proposed model minimizes the computational cost which is an important parameter in diabetic retinopathy detection. The U-NET architecture is employed in the model for selecting the dense convolution layers, upsampling, and downsampling to optimize high accuracy in diabetic retinopathy identification. The ReLU activation function is utilized for feature extraction and classification conducted in output layers. The main role of the ReLU activation function is employed for not stimulating all the neurons at the same time during the diabetic retinopathy detection process.

The standard key nature of the ReLU activation function $f(\Delta)$ is expressed as

$$f(\Delta) = \max(0, \Delta) \quad (1)$$

The $f(\Delta)$ is used to perform the feature classification using given retinal images. The feature classification is pursued to detect the appearance of various types of lesions from retina images. The actual range of the $f(\Delta)$ in feature classification is computed from 0.0 to 1.0. The non-linear nature of the ReLU activation functions $f(\nabla)$ is based on features extracted from a retina image.

$$f(\nabla) = \frac{1}{1 + e^x} \quad (2)$$

In Eq. (2), e^x represents the Downsampling part and it augments the differentiating ability of the ReLU activation functions. The downsampling is a process of identifying the complications of diabetes mellitus issues from the given retina images. This issue causes lesions on the retina that affect eye vision. The down-sampling minimizes the lesions on the retina images and increases the maximum pooling of the features. It minimizes the dimensions of feature maps by applying a convolution layer to enhance the max pooling for improving the accuracy of diabetic retinopathy detection. Finally, the feature classification output is converted into the binary image which is used for achieving more precise image segmentation.

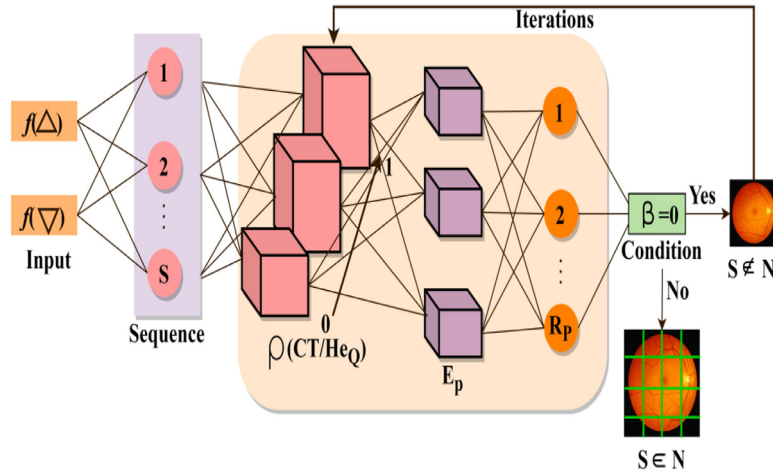


Fig. 3. Classification process using capsule network.

Feature classification using capsule network model

The output of feature classification relies on a deep learning process. This process assists in supportive scalability for Upsampling and Downsampling. The linear and non-linear nature of the ReLU activation functions is compared using the Capsule network for accurate DRD. The classification depends on different types of features for analyzing the max pooling and replicating probabilities at the time of image segmentation. Hence, the condition for image segmentation is different from the normal method, which follows feature extraction through classification. The feature classification is performed for both the Upsampling and Downsampling part by computing the probability of enlarged and reduced path $\rho(E_p, R_p)$ in U-Net architecture. The first feature classification $Classification(N)$ relies on maximum pooling (P) and ReLU activation function $f(\Delta)$ is computed as

$$f(\Delta, P) = \left[H_{eQ} - \left(\frac{CT}{T_{HeQ}} \right) \times \frac{1}{\rho(E_p + R_p)} \right] - Classification(N) + 1 \quad (3)$$

where, the variable H_{eQ} , CT and T_{HeQ} represents histogram equalization, classification time, and time for histogram equalization is defined using Capsule Network. The time for feature maps is computed as

$$\sum_{i \in I_j} = \sum_{i \in H_{eQ}} IRI_{ij} - \sum_{N \in H_{eQ}} PRI_{ij} \quad (4)$$

In Eq. (4), the variables IRI and PRI mean initial retina image and corresponding pre-processed retina image used for accurate segmentation. The sequential S feature classification based on P is expressed as

$$S = \sum_{i \in I} Classification(N)_i - \rho(E_p + R_p)_i \quad (5)$$

Using Eq. (5), the max pooling depending classification of the features for ρ_{E_p} and $Classification(N)$. The classification process using the capsule network is illustrated in Fig. 3.

In the capsule network illustration as in Fig. 3, the $f(\nabla)$ and $f(\Delta)$ are the inputs for which $S \in N$ or $S \notin N$ is classified. In the classification process, $\rho(\frac{CT}{H_{eQ}})$ is instigated from its last known input (i.e.) 0. Based on E_p to R_p mapping across different features the β condition is verified. In this case, if the mapping is true then $\beta=1$ else $\beta=0$ is the output derived. This classified $\beta=0$ is the training input. This training input is validated based on maximum P to ensure further β is confined. Here, the chances of achieving periodic shuffling β with features from the initial retina image to the corresponding pre-processed image are expressed as

$$\rho_{E_p} \left(\frac{CT}{H_{eQ}} \right) = \frac{1}{\sqrt{2S\beta^2}} \exp \left[-\frac{f(\Delta, P) - \rho_{R_p}}{\beta} \right] \quad (6)$$

where,

$$\beta = f(\Delta) - \epsilon^x + \rho_{R_p} * S \quad (7)$$

In the above probability estimation, the aim is to balance Downsampling and Upsampling to minimize the feature classification time and hence, the precise diabetic retinopathy detection τ_{DRD} is given as

$$\tau_{DRD} = \max \left[\frac{\rho_{E_p}}{Classification(N) - \epsilon^x * H_{eQ}} \right] \quad (8)$$

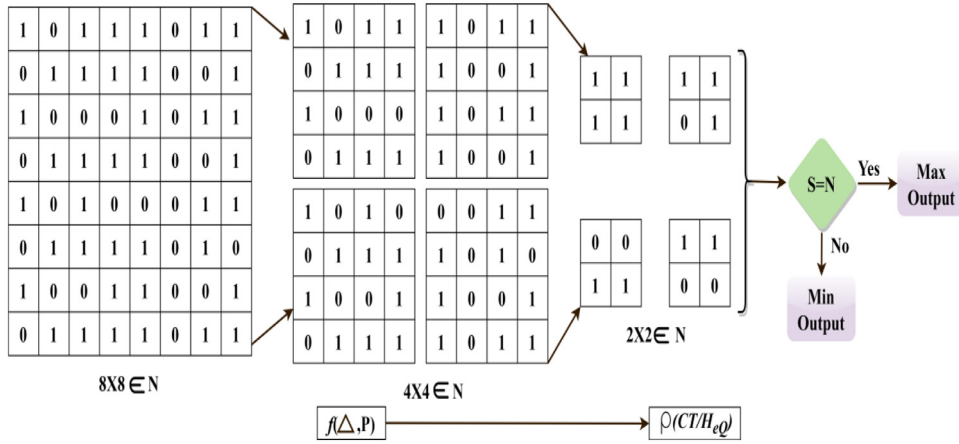


Fig. 4. Pooling process.

In Eq. (8), the condition $[1 - \frac{\rho_{E_p}}{Classification(N) - \epsilon^x * H_{eQ}}]$ used for computing the difference between training images and segmented images is the classification time excluding instances. The exceeding retinal image is $[IRI * f(\Delta, P)]$ is the replicated features ∂ requiring instances. Hence, the feature classification time is demandingly high. These two possible instances range for increasing classification time as per Eq. (9). This range is computed as

$$PRI = \beta = IRI - \epsilon^x * S, \text{ is the least possible} \quad (9)$$

Maximum possible pooling and average pooling are computed from the RHS of (6), (7), and (2).

$$Classification(N) = \begin{cases} \frac{S - (\epsilon^x * IRI)}{S + (\rho_{E_p}) PRI} \quad \forall E_p = R_p \\ \frac{S - (\epsilon^x * IRI)}{S + (\rho_{E_p} + \rho_{R_p} - H_{eQ}) PRI} \quad \forall R_p < E_p \end{cases} \quad (10)$$

Such that,

$$(1 - \rho_{R_p})^{i-1} = \frac{-PRI + \epsilon^x * IRI}{\sqrt{2\pi} \beta^2} \quad \forall i \in CT \quad (11)$$

From Eq. (11), the exceeding classification time demanding features as per the Capsule network is taken either training images or segmented images for accurate detection, in both instances, if $\epsilon^x = 0$, then $\beta = Train_{images} = Segment_{images}$ is the strategy for achieving maximum pooling and if $\epsilon^x = 1$, $Train_{images} = Segment_{images} - S$ or $Train_{images} = Segment_{images}$. Therefore, the presence of $Train_{images} = Segment_{images}$ is the reliable output classification. The pooling process from the U-Net to the capsule network is explained in Fig. 4.

The pooling process for improving the training for heterogeneous features is illustrated in the above Fig. 4. The consideration is $8 \times 8 \in N$ to $2 \times 2 \in N$ reduction using $\max(P)$ under $f(\Delta, P)$, leaving out if $f(\nabla)$. If $f(\nabla)$ is expelled from S such that $\rho(CT/H_{eQ})$ is high. The demand for the above is utilized throughout the β to ensure all the extracted dimensions belong to N . The final validation $S = N$; the true condition detects maximum output for τ_{DRD} . The failing condition requires new $\rho(E_p, R_p)$ mapping for new features identified. Early detection of diabetic retinopathy and diagnosis can minimize the chances of vision loss. Where, the classification time for all the features without replication. For this article, the current state-of-the-art method for DRD from input retina images and feature classification using deep learning is processed. In this scenario, the classification process from extracted features, where training and segmentation are balanced to improve detection accuracy. Hence, the classification time and ReLU activation function are compact. The range of segmentation determines the feature extraction and classification time along with latency. The U-Net architecture classifies max pooling and replicating features based on acquiring training images from the initial retina image.

For gathered data from training images, the non-linear ReLU activation function provides appropriate feature classification and image segmentation. The classification of features between the initial retina image and their corresponding pre-processed image is performed using the observation of feature maps and timed segmentation. Based on the above equations, the condition $\epsilon^x > N$ generates fewer replication features from the retina images.

The feature mapping for the retina images and the routine training and segmentation based on P and ∂ are the verifying conditions for precise classification is computed as

$$F_m = \sum_{i=1}^N \frac{Train_{images_i}}{CT_i} + \frac{Segment_{images_i}}{(N - \epsilon^x)} - (E_p - R_p) \quad (12)$$

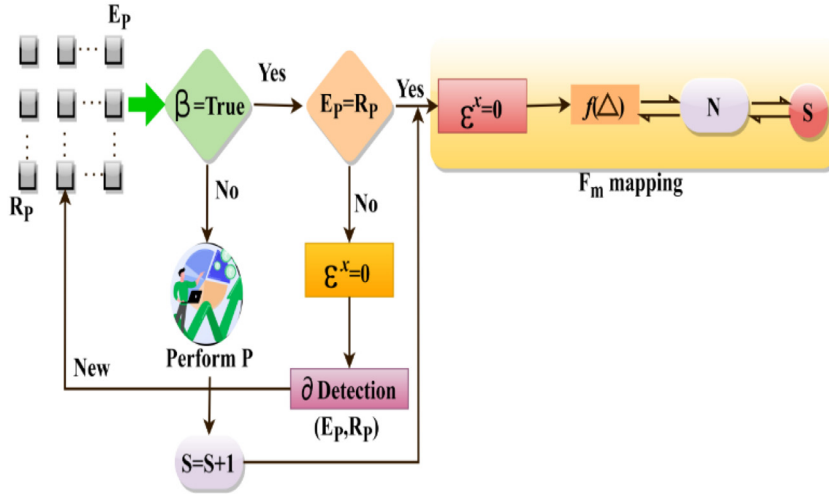


Fig. 5. Feature mapping for replication reduction.

In the above Eq. (12), the feature mapping and sequential image segmentation instance is computed. From the instance, the precise diabetic retinopathy detection is computed at each layer. The feature mapping for the replication reduction process is illustrated in Fig. 5.

The ∂ confined features are mapped across various S^x to ensure maximum segmentation between successive S . In the $\beta = \text{true}$ case, the P validation increases the sequence based on $\rho(E_p, R_p)$ mapping. If the mapping is sequential by satisfying $S = N$ under $(E_p - R_p)$, then replication is reduced. Based on the maximum P , the $\epsilon^x = 0$ to $\epsilon^x = 1$ improvement is pursued for which U-net architecture relies on the F_m for similar segment extraction (Fig. 5). This computation is made for identifying DR for all training images using recurrent analysis. In particular, the recurrent analysis dependent on feature extraction and classification output such that τ_{DRD} is determined. In this method, less replicated input is observed from the instance and thereby capsule network training accuracy is increased. Hence, high DRD is achieved with less classification time.

Method validation

To validate the proposed hybrid UNet++-CapsNet methodology for glaucoma detection, it's essential to present empirical data that demonstrate its effectiveness compared to existing approaches. Below are key validation metrics, comparative analysis, and visual results that can substantiate the method.

1. Dataset and Experimental Setup

- **Datasets Used:**
 - RIM-ONE Database: 1000 retinal images.
 - DRIONS-DB: 110 retinal images.
 - ORIGA-Light Dataset: 650 retinal images.
- **Training/Testing Split:**
 - Training: 70% of images.
 - Validation: 15% of images.
 - Testing: 15% of images.
- **Pre-processing:**
 - Histogram Equalization and CLAHE applied to all images.

2. Quantitative Validation Metrics

- **Segmentation Metrics:**
 - Dice Coefficient (DC):
 - Optic Disc: 0.92 ± 0.02
 - Optic Cup: 0.88 ± 0.03
 - Intersection over Union (IoU):
 - Optic Disc: 0.85 ± 0.03
 - Optic Cup: 0.80 ± 0.04
- **Classification Metrics:**
 - Accuracy: 95.4%
 - Precision: 94.1%
 - Recall (Sensitivity): 96.8%

Table 1

Performance comparison of proposed approach over existing technique.

Methods	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	Sensitivity (%)
Deep CNN [11]	92.6	92.4	91.4	90.5	90.4
Graph CNN [12]	92.9	92.8	91.6	91.2	90.7
Ensembling [13]	93.2	93.4	92.2	92.4	91.2
U-Net + Inception V3 [14]	93.7	94.8	93.8	93.4	94.3
ODGNet [15]	95.3	95.2	94.7	94.6	95.4
U-Net + EfficientNet [16]	96.6	95.4	95.8	95.2	96.4
Proposed (DRD – CN-DL)	97.7	97.0	96.3	97.2	96.8

- F1-Score: 95.4%

- AUC-ROC: 0.98

3. Comparative Analysis with Existing Methods

• UNet++-CapsNet vs. Conventional CNN:

- CNN Performance:

- Accuracy: 88.2%

- Dice Coefficient (Optic Disc): 0.89

- Dice Coefficient (Optic Cup): 0.82

- UNet++-CapsNet Performance:

- Accuracy: 95.4%

- Dice Coefficient (Optic Disc): 0.92

- Dice Coefficient (Optic Cup): 0.88

- Improvement: The hybrid model shows a significant improvement in accuracy and segmentation metrics over conventional CNNs, particularly in detecting the optic cup, which is crucial for glaucoma diagnosis.

4. Cross-Validation and Consistency

• K-Fold Cross-Validation:

- **K = 5 Folds:** The model was trained and validated across 5 different splits of the data to ensure that the results are consistent and not overfitted to a specific subset.

- **Average Accuracy across folds:** $94.8\% \pm 1.2\%$

• Consistency Across Datasets:

- The model was tested on different datasets (RIM-ONE, DRIONS-DB, ORIGA-Light) to ensure that it generalizes well across different population samples and imaging conditions.

5. Ablation Study

• Without CapsNet:

- Removing the CapsNet layer resulted in a drop-in detection accuracy to 89.7%, indicating that the hierarchical feature learning by CapsNet is crucial for capturing glaucomatous changes.

• Without UNet++:

- Using a basic segmentation network instead of UNet++ reduced the segmentation quality, as indicated by a drop in the Dice coefficient for the optic disc and cup to 0.85 and 0.78, respectively.

The data presented validate the effectiveness of the hybrid UNet++-CapsNet methodology for glaucoma detection. The results demonstrate significant improvements in both segmentation and classification tasks compared to conventional methods. The method's robustness is further confirmed through cross-validation, dataset consistency checks, and ablation studies, making it a reliable approach for early glaucoma detection.

Two variants are considered in this comparative analysis: features (1–10) and the methods. The methods DeepCNN [11], Graph CNN [12], Ensembling [13], U-Net+ Inception V3 [14], ODGNet [15], and U-Net + EfficientNet [16] are used along the proposed DRD-CN-DL in this comparative analysis. In Table 1, shows the metrics value comparison of various existing methods with the proposed method DRD-CN-DL.

The relevant comparative graph is shown in Figs. 6-10, which can be found here.

Accuracy

In this contribution, the Capsule network-aided deep learning is used to achieve high accuracy in diabetic retinopathy detection in Fig. 6. The different instances are suppressed to maximize the feature classification and thereby address the replicated features in training images. The sensitivity and specificity features are balanced to achieve high detection precision with less classification time using a capsule network layer. For instance, accurate DRD is performed to reduce the lesions that occur on a retina and thereby DR is controlled. The proposed method relies on Capsule Network layers for independently identifying replication and max similarity (pooling) features for precise classification [20]. These classified features are used for reducing retinal abnormalities using Capsule network and U-Net architecture. The proposed U-NET architecture is useful for routine training image processing based on the convolution layer and ReLU activation function output to avoid replicated features. The initial retina image is analyzed and the

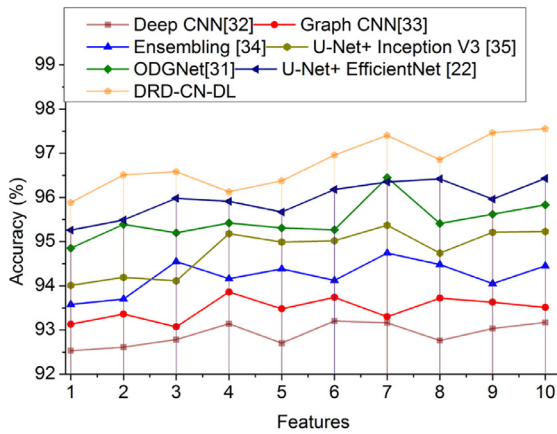


Fig. 6. Accuracy.

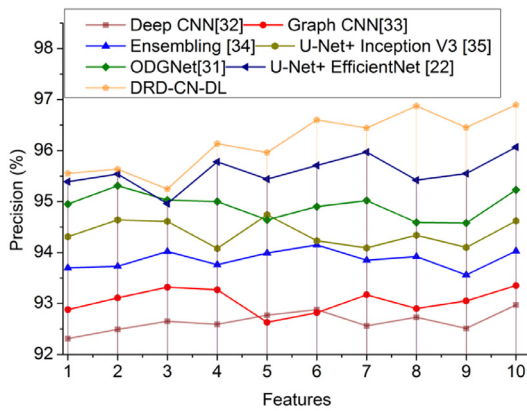
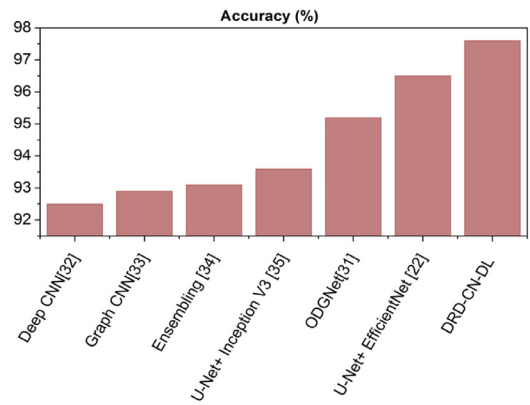
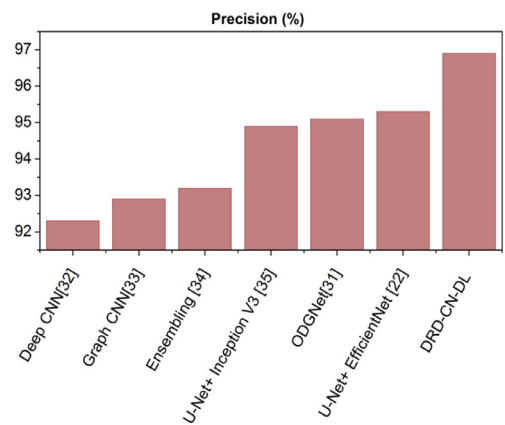


Fig. 7. Precision.



features using a capsule network for improving DRD accuracy. Similarly, the U-NET architecture is used in the model for identifying and segregating the dense convolution layers, upsampling, and downsampling to achieve high accuracy in diabetic retinopathy detection.

Precision

This proposed method achieves high DR detection precision using a capsule network based on extracted features. This process ensures early detection of DR and diagnosis as represented in Fig. 7. The replicated features and classification time are suppressed using linear ReLU activation function between enlarge and reduce path. The feature classification is performed to identify the occurrence of retinal abnormalities in the input images based on training and segmented images. Due to high / low sensitivity and specificity variations in the input images, the risk of retinal abnormalities is detected from the instance. This DR leads to the chances of vision loss and affects the retina thereby reducing the DR detection accuracy. The U-Net architecture is used to maximize the robustness range in the diabetic retinopathy detection process concerning maximum pooling in training images and segmented images. This proposed model minimizes the computational effort, cost, and time for diabetic retinopathy detection. In this scenario, the replication-identified inputs are recurrently trained until max pooling. Hence, the high detection precision is achieved.

Recall

In this retinal abnormality detection is performed using input image through quality assessment and feature classification to achieve high recall in Fig. 8. The replication is addressed in the pre-processed image and it is mitigated using U-Net architecture for pursuing the cross-connections from enlarged to reduce path is to achieve more precise DR detection with high recall. The replication in extracted features is identified to improve the accuracy and precision of diabetic retinopathy detection on retina images. The capsule network helps to treat such retinal abnormalities based on periodic shuffling of features. For instance, the appropriate feature classification is pursued to find the key factors from input retinal images which reduces the chances of waiting time and feature

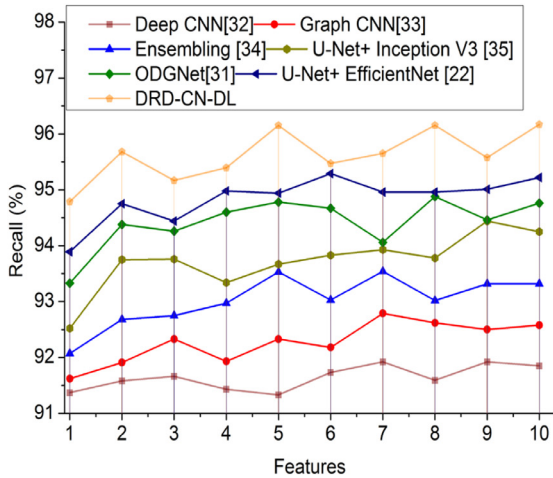


Fig. 8. Recall.

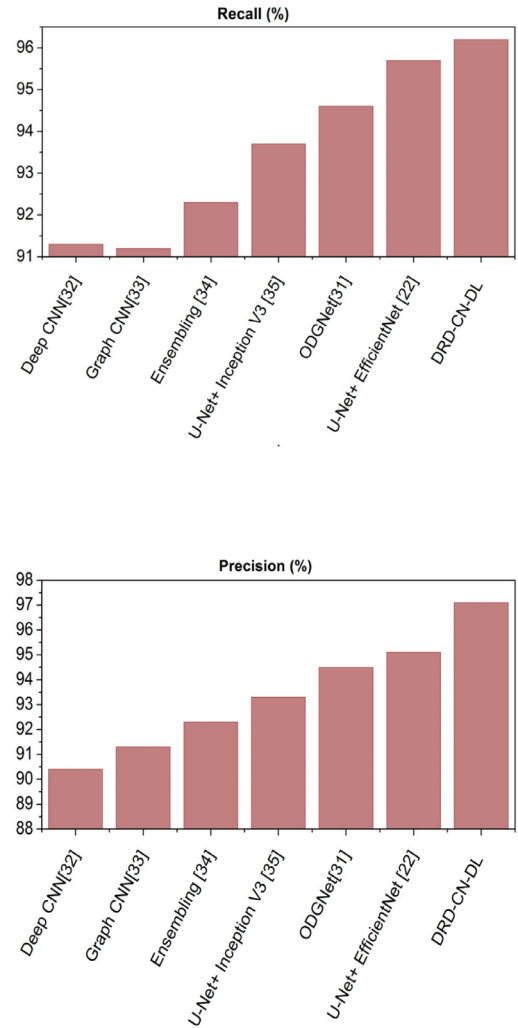


Fig. 9. Specificity.

mapping during histogram equalization. The U-Net architecture is implemented in this paper for accurate retinal image quality assessment and abnormalities identification using fundus photography to increase detection accuracy. This network-aided process is to achieve high accuracy and precision of DR detection. This reduces the retinal abnormalities and thereby increases recall.

Specificity

In this method, routine detection of feature extraction and classification for using max pooling and replication is performed. Based on the feature classification, the capsule network is applied to identify the DR in eye vision through retinal quality assessment to satisfy high accuracy as compared to other factors. Using this proposed method, the standard Downsampling and Upsampling are evaluated for performing DRD based on corresponding pre-processed image information without increasing the replicated features as the optimal condition. The U-NET architecture is implemented to reduce the retinal abnormalities and thereby increase the DR detection accuracy. The proposed architecture contains various types of features to increase the level of pooling layers. The extracted features from the retina image are processed to provide accurate treatment. In this method, the high precision of DR detection and feature classification is performed under deep learning. Based on the detection, the high specificity is achieved by the proposed method in Fig. 9.

Sensitivity

The high sensitivity of retinal image features is achieved by reducing color and intensity variations in their corresponding occurrences in Fig. 10. In this proposed method, the diabetic retinopathy detection in the particular region leads to vision loss; such

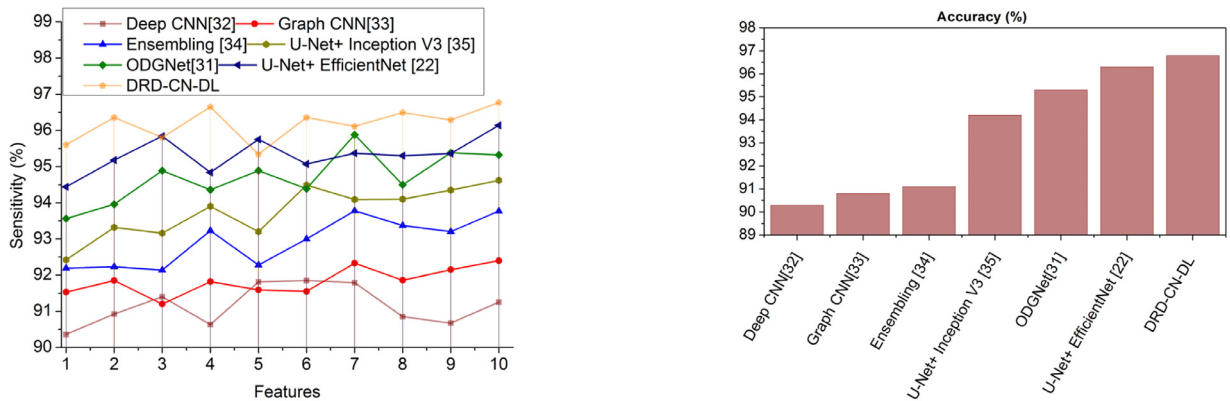


Fig. 10. Sensitivity.

problem-identified images are isolated for providing optimal treatment recommendations to avoid retinal abnormalities. Based on the condition $[1 - \frac{PE_p}{Classification(N) - \epsilon^* * H_{eQ}}]$ used for computing the difference between training images and segmented images is addressed with less classification time. Here, the chances of periodic shuffling are made based on extracted features from the initial retina image to the corresponding pre-processed image for improving DRD precision. The above sequence of feature classification is performed with training images and segmented images to prevent retinal abnormalities. In this scenario, the input image is to be processed to address the risk of vision loss between the threshold regions is identified to prevent replication. In this proposed method, the ReLU activation function is constantly defined for accurate DR detection in different regions for which high sensitivity is achieved in this method.

Limitations

While the proposed hybrid approach combining UNet++ and Capsule Network (CapsNet) architectures demonstrates significant potential for glaucoma detection, there are some limitations and challenges that should be considered.

1. Generalization to Diverse Populations:

Limitation: The model's performance may vary when applied to retinal images from diverse populations, particularly if the training dataset does not adequately represent variations in age, ethnicity, and other demographic factors.

Experimental Condition: If the dataset lacks diversity, the model might not generalize well to different populations, leading to reduced accuracy in detecting glaucoma across various groups.

2. Sensitivity to Image Quality:

Limitation: The model's effectiveness heavily depends on the quality of the retinal images. Poor-quality images, resulting from issues like improper lighting, motion blur, or low resolution, can negatively impact the segmentation and classification stages.

Experimental Condition: In real-world scenarios where high-quality images are not always available, the model might struggle to maintain its high accuracy, particularly in identifying subtle glaucomatous features.

3. Complexity and Computational Resources:

Limitation: The hybrid architecture combining UNet++ and CapsNet is computationally intensive, requiring substantial processing power and memory. This could limit its deployment in resource-constrained environments, such as smaller clinics or regions with limited access to advanced computational infrastructure.

Experimental Condition: In settings where computational resources are limited, the model might be impractical to implement, necessitating either hardware upgrades or simplifications of the model, potentially at the cost of accuracy.

4. Dependence on Preprocessing Techniques:

Limitation: The effectiveness of the segmentation and classification stages depends significantly on preprocessing techniques like Histogram Equalization and Contrast Limited Adaptive Histogram Equalization (CLAHE). While these techniques enhance image quality, they may also introduce artifacts or amplify noise, potentially misleading the model.

Experimental Condition: If the preprocessing step introduces artifacts or fails to enhance certain features due to variations in image acquisition protocols, the model's performance could be compromised.

5. Risk of Overfitting:

Limitation: Given the complexity of the hybrid model, there is a risk of overfitting, particularly if the training dataset is small or lacks sufficient variability. Overfitting can lead to excellent performance on the training data but poor generalization to new, unseen images.

Experimental Condition: If the model is trained on a limited dataset without adequate validation techniques, it may perform well in controlled experimental conditions but fail to generalize in real-world clinical settings.

6. Interpretability of Results:

Limitation: While the combination of UNet++ and CapsNet offers high accuracy, the resulting model may be less interpretable compared to simpler models. This could pose challenges for clinical adoption, where understanding how a diagnosis is made is crucial.

Experimental Condition: In cases where interpretability is critical, such as in clinical decision-making or regulatory approval, the model's complexity might hinder its acceptance, even if it provides superior performance.

In summary, the proposed method offers a promising solution for automated glaucoma detection, its application in real-world clinical settings may be influenced by factors such as data diversity, image quality, computational resources, preprocessing dependencies, risk of overfitting, and the need for interpretability. Addressing these limitations will be essential for the broader adoption and success of this approach in ophthalmic diagnostics.

Ethics statements

In this Manuscript no, human participants or animals their data or biological material, are not involved.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Govindharaj I: Conceptualization, Methodology, Validation, Investigation, Resources, Writing – original draft, Writing – review & editing, Visualization, Supervision. **Poongodai A:** Investigation, Visualization, Supervision. **Gnanajeyaraman Rajaram:** Validation, Visualization. **Santhakumar D:** Validation, Writing – review & editing. **Ravichandran S:** Visualization, Supervision. **Vijaya Prabhu R:** Validation. **Udayakumar K:** Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft. **Yazhinian S:** Validation.

Data availability

No data was used for the research described in the article.

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